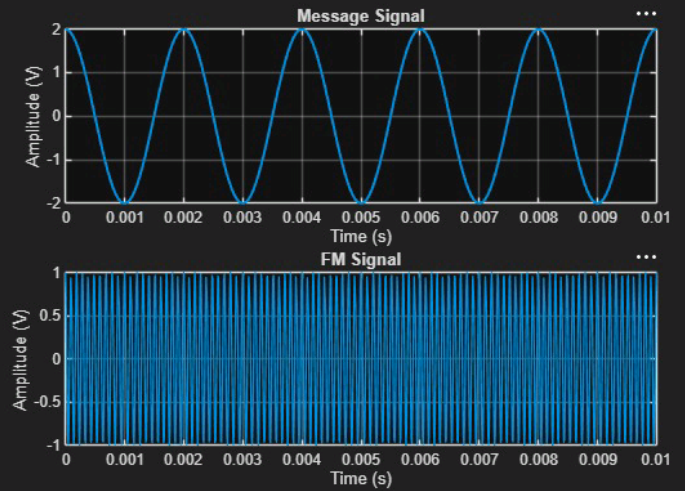


> MATLAB Drive

/MATLAB Drive/experimentadc22.m

```
30 fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
31 fprintf('Modulation Index (Beta) = %.2f\n',beta);
32 %% Verification
33 beta_verify = delta_f/fm;
34 fprintf('Verified Modulation Index = %.2f\n',beta_verify);
35 %% Verification Check
36 tolerance = 1e-6;
37 if abs(beta - beta_verify) < tolerance
38     fprintf('\nVerification Successful\n');
39 else
40     fprintf('\nVerification Failed\n');
41 end
42 %% Plotting
43 figure;
44 subplot(2,1,1);
45 plot(t,m,'LineWidth',1.5);
46 grid on;
47 title('Message Signal');
48 xlabel('Time (s)');
49 ylabel('Amplitude (V)');
50 subplot(2,1,2);
51 plot(t,fm_signal,'LineWidth',1);
52 grid on;
53 title('FM Signal');
```

Figure 1 ×



on (Delta f) = 500.00 Hz  
(Beta) = 1.00

